**Highlighted component indicates the custom wrist-roll joint addition.*

ur5

6-DOF Upgraded Edition

Full Orientation. Designed for research and educational applications.

The upgraded ur5 features a custom 6-DOF configuration with an integrated wrist-roll joint, enabling full spatial orientation. With a compact footprint and an open-source control stack, it provides a cost-effective solution for robotic manipulation, rapid prototyping, and educational applications.

Key Features

- Full 6-DOF spatial orientation capability
- Native out-of-the-box ROS 2 integration
- Lightweight, modular, and upgradeable design
- Built-on industrial Dynamixel smart servos
- Cost-effective platform for advanced R&D

Modifications vs. Original

- **Hardware:** Added 1 DOF (wrist-roll joint)
- **Actuation:** J5 utilizes XL430-W250-T servo
- **Software:** Updated URDF & MoveIt kinematics

Applications

- Pick-and-place automation testing
- Algorithm benchmarking & motion planning
- University robotics education and training

Performance

Max TCP speed	0.30 m/s
Max joint speed	275°/s
Mounting Options	Tabletop or fixed base
Noise Level	¡ 60 dB(A)
Orientational Accuracy	± 0.663 °
Positional Accuracy	± 0.545 mm

General Specification

Payload	0.3 kg (at max reach)
Reach	846 mm
Degrees of Freedom	6
Configuration	Articulated Manipulator
Repeatability	±2.067 mm
Footprint	Ø150 mm

*Payload includes end-effector weight. Reach defined as maximum distance from base to TCP.

Joint Specifications

Axis	Working Range	Max Speed	Peak Torque
Shoulder Pan Axis	$-360°$ to $360°$	180°/s	4.1 Nm
Shoulder Lift Axis	$-360°$ to $360°$	180°/s	4.1 Nm
Elbow Axis	$-180°$ to $180°$	180°/s	4.1 Nm
Wrist 1 Axis	$-360°$ to $360°$	180°/s	4.1 Nm
Wrist 2 Axis	$-360°$ to $360°$	180°/s	4.1 Nm
Wrist 3 Axis	$-360°$ to $360°$	180°/s	4.1 Nm

*Gripper provides additional actuation DOF. Torque values are derived from manufacturer datasheets.

Control System

Supported OS	Ubuntu 22.04 / 24.04
Middleware	ROS 2 (Humble / Jazzy)
Motion Planning	MoveIt 2 (OMPL)
Controller	ros2_control
Simulation	Gazebo (Ignition)
Programming	Python, C++ APIs
Interface	USB / Serial (via controller)

Actuators

Actuator Type	DYNAMIXEL Smart Servo
Motor Models	XM430-W350-T / XL430-W250-T
Control Mode	Position / Velocity / Current
Communication	TTL Multi-Drop Bus
Feedback	Absolute Contactless Encoder

Electrical Specifications

Input voltage	12 V DC
Power supply	12 V, 5 A adapter
Typical power	35 W
Maximum power	120 W (peak load)
Digital I/O	Available via external controller

Physical & Kinematics

Materials	PLA / Aluminum
Cable length	1 m
Total Weight	$\approx$ 21.05 kg
IP Rating	IP20
Operating Temp	0°C to 45°C
Humidity	5% to 85% (non-condensing)

Safety & Compliance

Safety rating	Uncertified (Research use only)
Architecture	Software-limited force/torque feedback (PL-b equivalent)
Protection	Configurable joint, velocity, and torque limits
E-Stop	Software-triggered (external hardware optional)

Known Limitations

- Not suitable for industrial payloads or high-cycle loads
- Limited stiffness due to 3D-printed PLA structure
- Accuracy subject to standard Dynamixel gear backlash

Documentation & Resources

GitHub Repository
Access source code, ROS 2 packages, setup guides, and updates.
github.com/KMTI-ROBOPARADIGM/omx-wrist-roll-original

Package Contents

- 6-DOF Robotic Arm & Gripper
- 12V Power Adapter
- Communication Cable
- Mounting Hardware
- Quick start guide / documentation

Contact & Origin

Lab/Entity	**AutoCert Lab**
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